

6.07 Reciprocal Functions

- 1** For each function, sketch the graph of the reciprocal function $y = \frac{1}{f(x)}$ and find its domain and range.

a $f(x) = x - 2$

b $f(x) = 2x + 6$

c $f(x) = x^2 - 4$

d $f(x) = x^2 - 2x - 8$

e $f(x) = (x + 1)^2$

f $f(x) = -x^2 - 9$

g $f(x) = x^3$

h $f(x) = x(x - 2)(x + 2)$

i $f(x) = (x + 1)(x + 3)(x - 2)$

j $f(x) = -x(x - 1)^2$

- 2** Sketch the graph of the reciprocal function of:

a $f(x) = -5x - 2$

b $f(x) = x^2$

c $f(x) = x^2 + 4x + 3$

d $f(x) = 1 - x^2$

e $f(x) = x^2 + 3$

f $f(x) = -x^3$

g $f(x) = x^3 - 1$

h $f(x) = -x^3 - 8$

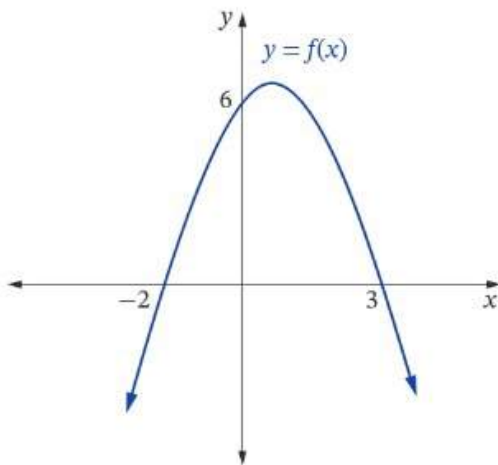
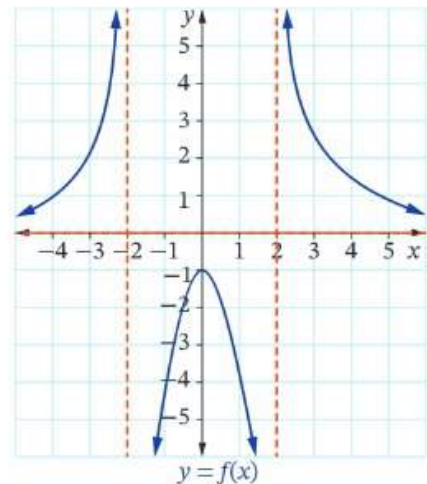
i $f(x) = -(x - 2)(x + 4)(x - 1)$

j $f(x) = (x - 3)(x + 1)^2$

- 3** If $y = f(x)$ is even, show that $y = \frac{1}{f(x)}$ is even.

- 4** If $y = f(x)$ is odd, show that $y = \frac{1}{f(x)}$ is odd.

- 5** For each graph of $y = f(x)$, sketch the graph of $y = \frac{1}{f(x)}$.

a**b**

6.08 Further Absolute Value Functions

- 1 For each function, sketch the graph of:

i $y = |f(x)|$

ii $y = f(|x|)$

a $f(x) = x + 5$

b $f(x) = 3x - 6$

c $f(x) = x^2 - 4$

d $f(x) = x^2 + 4x + 4$

e $f(x) = x^2 + x - 12$

f $f(x) = (x - 3)(x - 5)$

g $f(x) = x^3 + 1$

h $f(x) = (x - 2)(x + 4)(x - 1)$

i $f(x) = x(x - 3)^2$

j $f(x) = x^2(x + 1)$

- 2 Sketch the graph of $y = |f(x)|$ if $f(x) = x^2 + x - 6$, and state its domain and range.

- 3 For the function $f(x) = x + 4$, sketch the graph of:

a $y = \frac{1}{f(x)}$

b $y = f(|x|)$

c $y = |f(x)|$

- 4 If $f(x) = x^2 - 1$ and $g(x) = 3x - 3$, sketch the graph of:

a $y = f(|x|)$

b $y = |g(x)|$

c $y = \frac{1}{g(x)}$

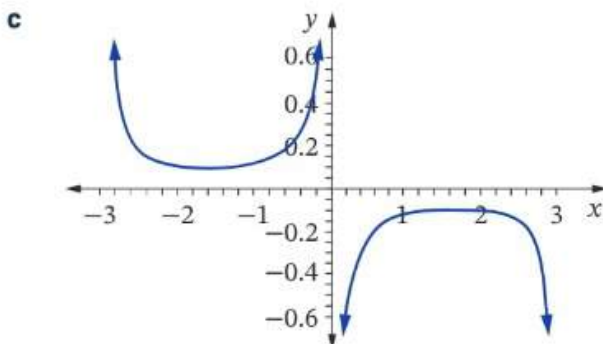
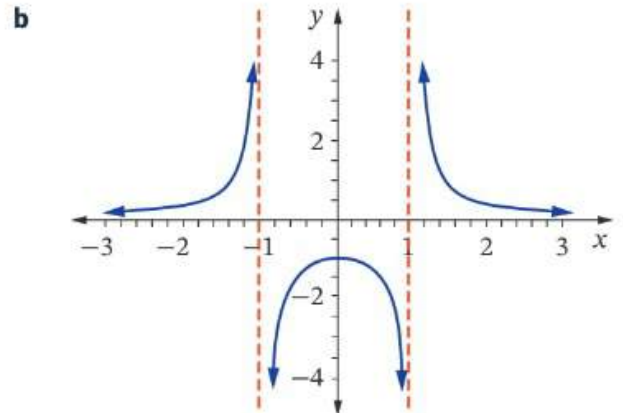
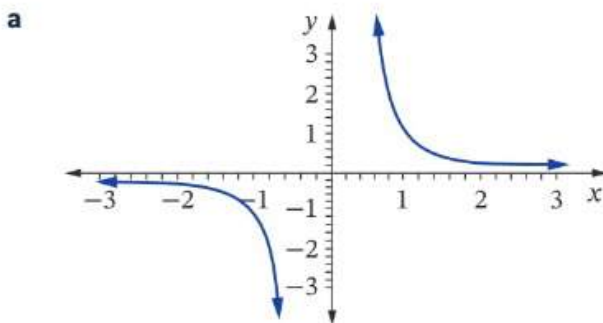
d $y = g(|x|)$

e $y = |f(x)|$

- 5 For each graph of $y = f(x)$, sketch the graph of:

i $y = |f(x)|$

ii $y = f(|x|)$



6.09 Sums and Differences of Functions

1 Sketch the graph of $y = f(x) + g(x)$ given:

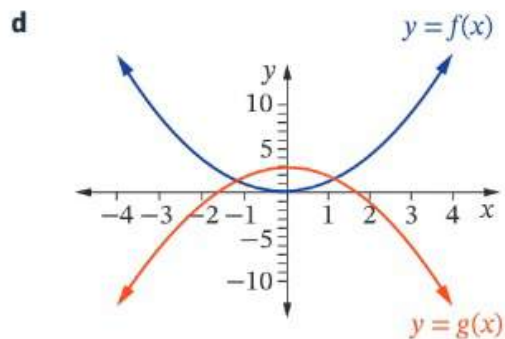
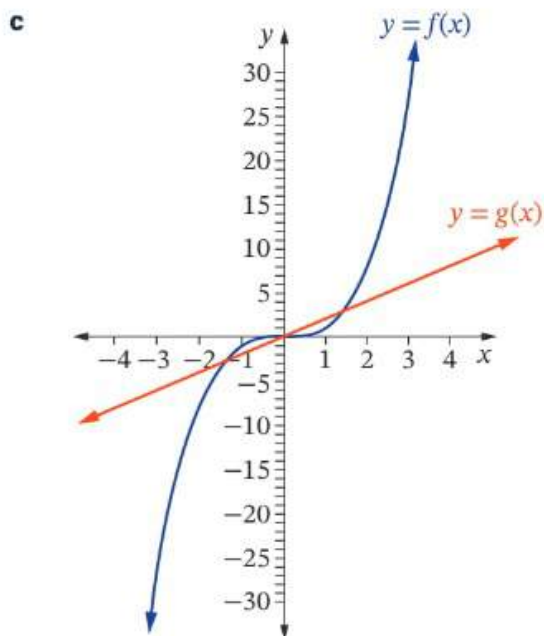
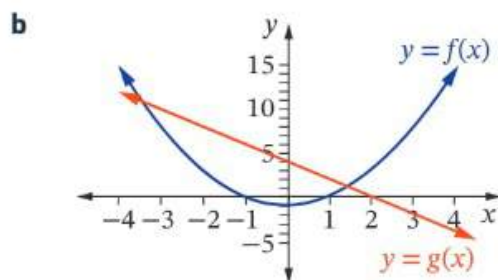
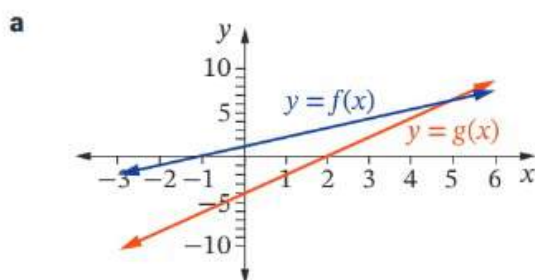
- a** $f(x) = 3x + 5$ and $g(x) = x - 1$ **b** $f(x) = x - 2$ and $g(x) = x^2$
c $f(x) = 3x$ and $g(x) = x^2 + x$ **d** $f(x) = -x - 3$ and $g(x) = -x^2 - 3x - 1$
e $f(x) = 3x - 4$ and $g(x) = x^2 + 2x - 2$ **f** $f(x) = -x^2 - 5$ and $g(x) = 2x + 1$
g $f(x) = -x + 1$ and $g(x) = x^3 + x + 7$ **h** $f(x) = x^3 - 4x - 1$ and $g(x) = x^2 - 3$

2 Sketch the graph of $y = f(x) - g(x)$, if:

- a** $f(x) = x + 1$ and $g(x) = x + 5$ **b** $f(x) = x - 3$ and $g(x) = 2x + 4$
c $f(x) = x^2 - 1$ and $g(x) = x + 3$ **d** $f(x) = x^2$ and $g(x) = x - 2$
e $f(x) = x + 4$ and $g(x) = x^2 - 6x + 5$ **f** $f(x) = -x^2$ and $g(x) = x^2 - 2x - 8$

3 For each graph of $y = f(x)$ and $y = g(x)$ shown, sketch the graph of:

- i** $y = f(x) + g(x)$ **ii** $y = f(x) - g(x)$

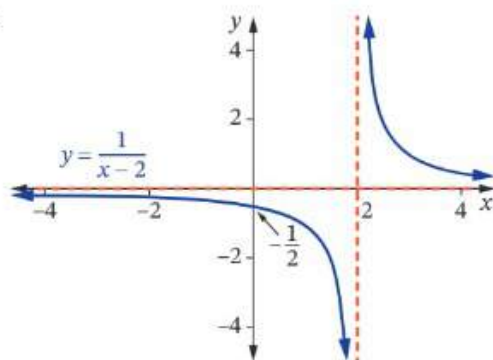


4 For the functions $f(x) = x^3 + 1$ and $g(x) = x^2 - 3$, sketch the graph of each new function and find its domain and range.

- a** $y = f(x) + g(x)$ **b** $y = f(x) - g(x)$

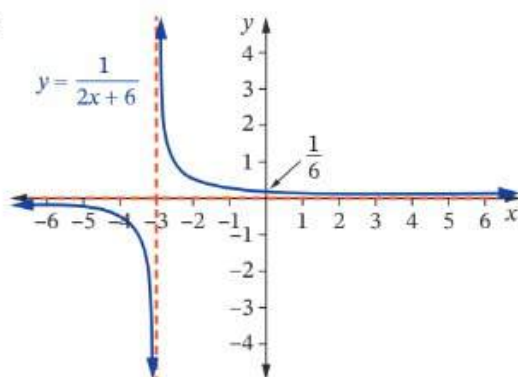
Exercise 6.07

1 a



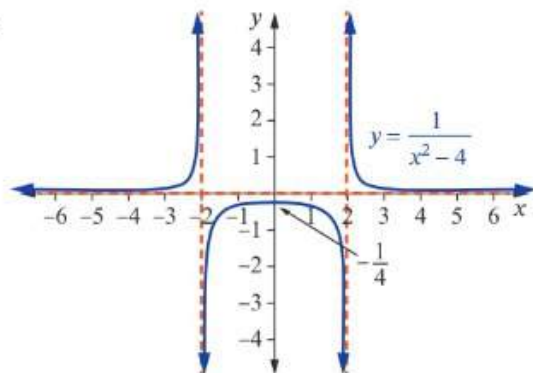
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b



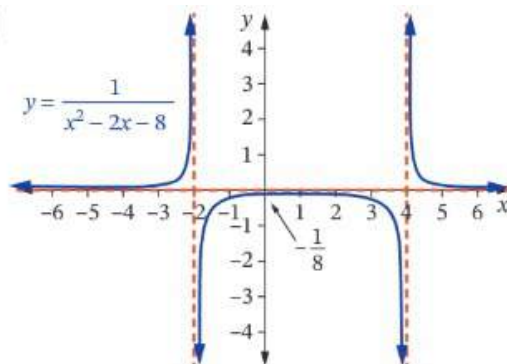
domain: all real x , $x \neq -3$,
range: all real x , $x \neq 0$

c



domain: all real x , $x \neq -2, 2$,
range: $y \leq -\frac{1}{4}$, $y > 0$

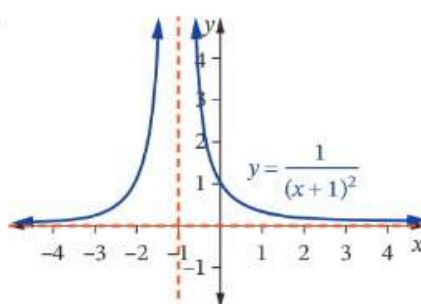
d



domain: all real x , $x \neq -2, 4$,

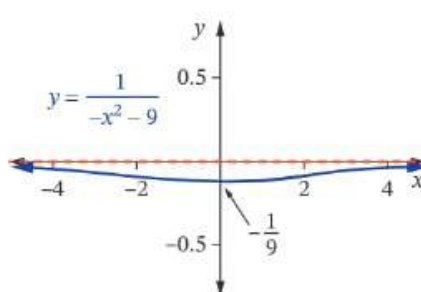
range: $y \leq -\frac{1}{9}$, $y > 0$

e



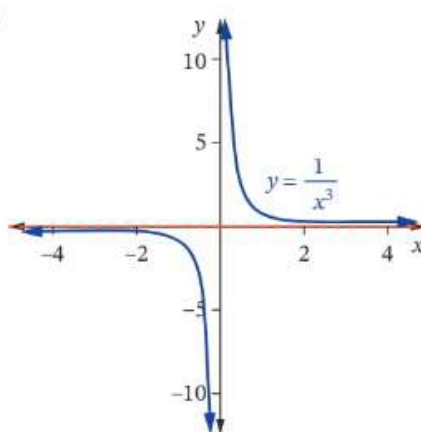
domain: all real x , $x \neq -1$, range: $y > 0$

f

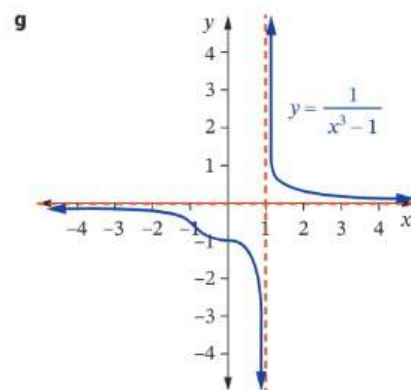
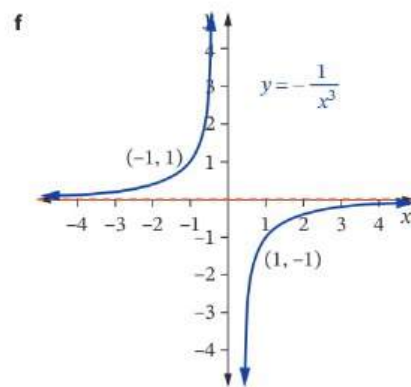
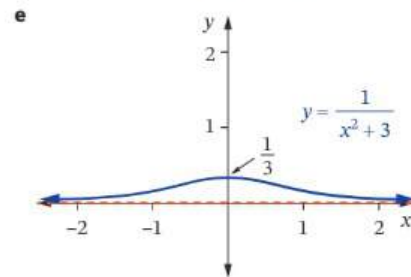
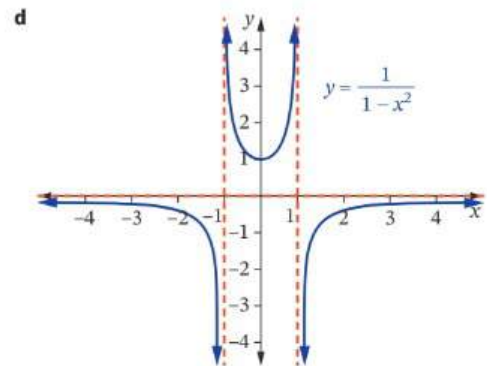
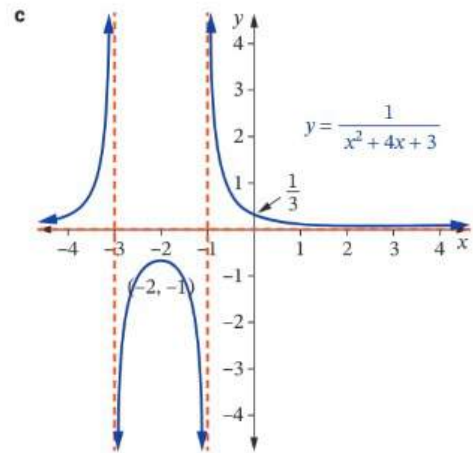
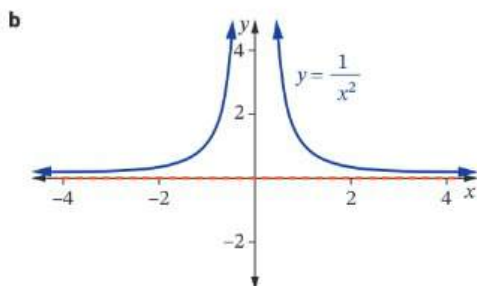
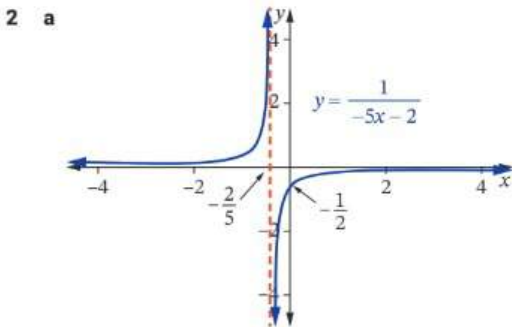
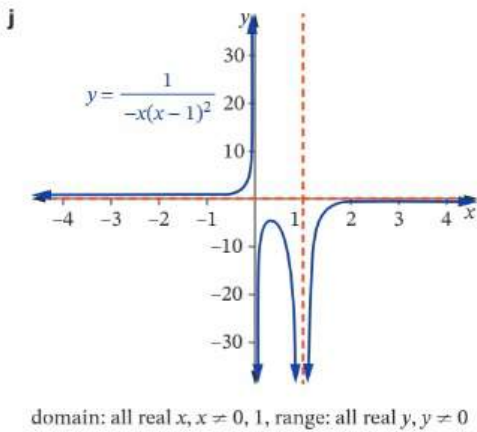
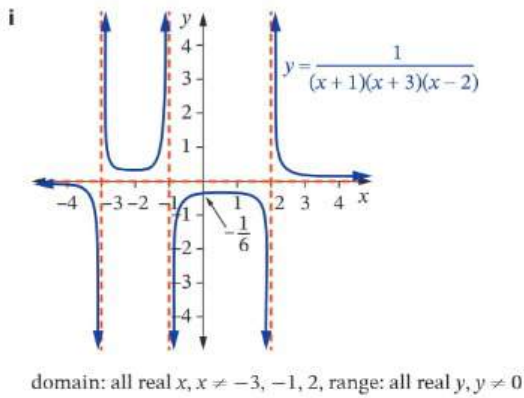
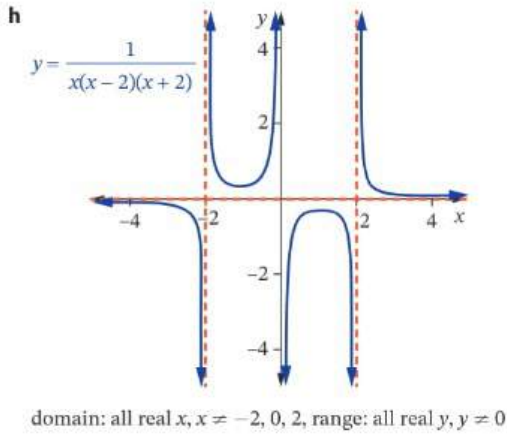


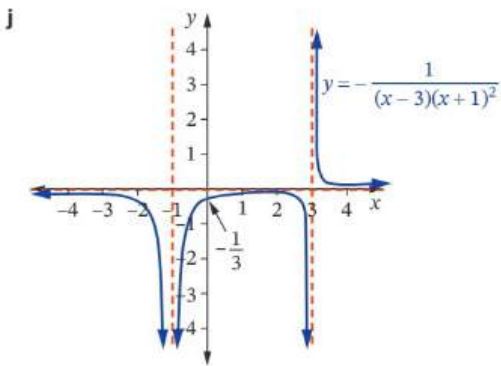
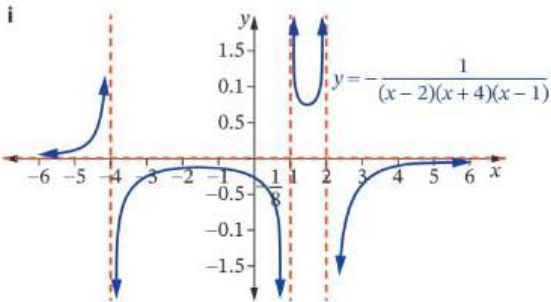
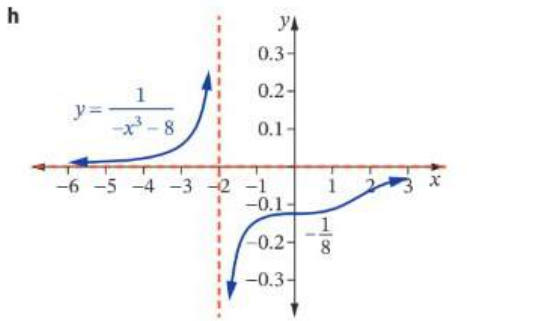
domain: all real x , range: $\left[-\frac{1}{9}, 0\right)$

g

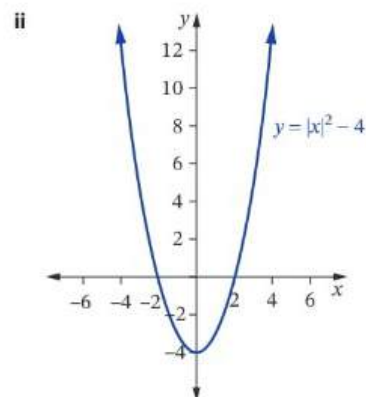
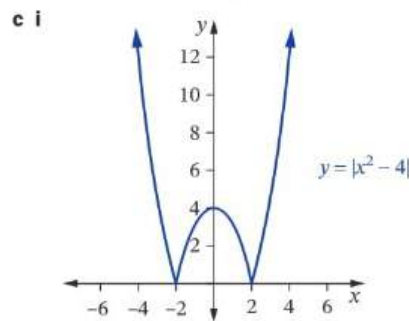
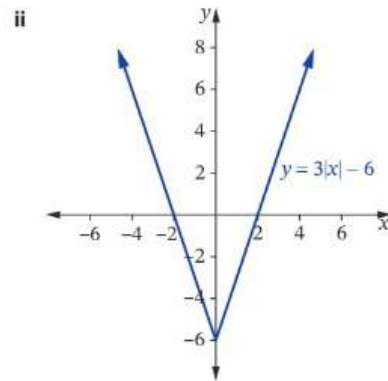
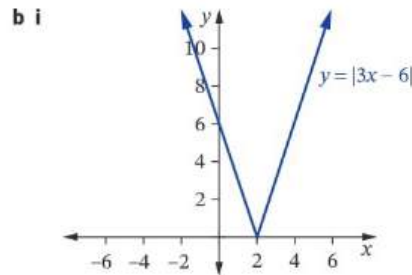
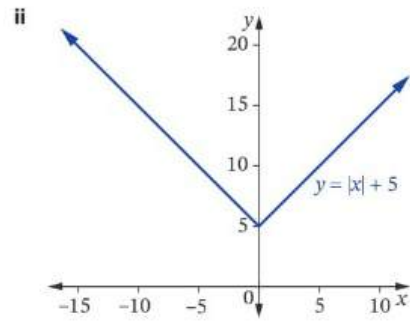
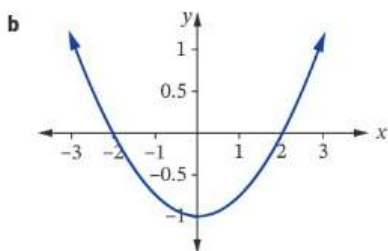
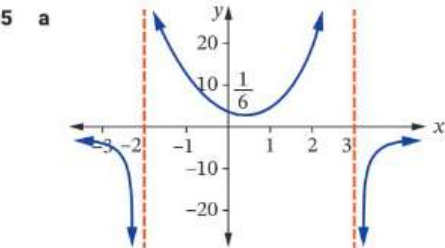


domain: all real x , $x \neq 0$, range: all real y , $y \neq 0$

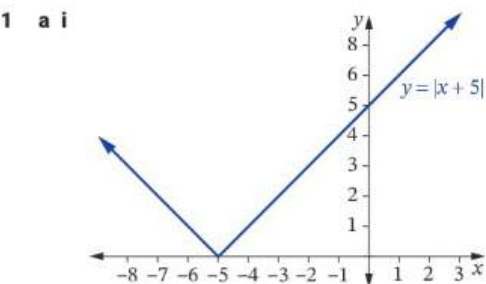




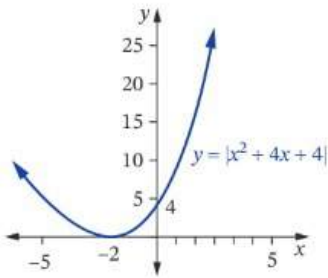
3-4 See worked solutions.



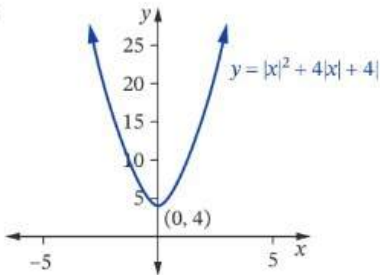
Exercise 6.08



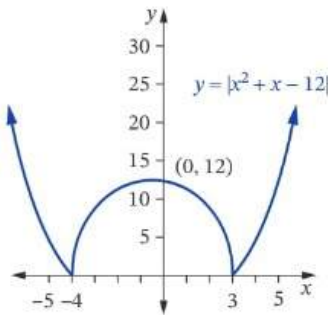
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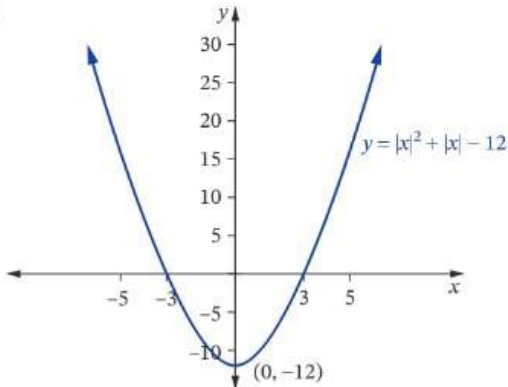
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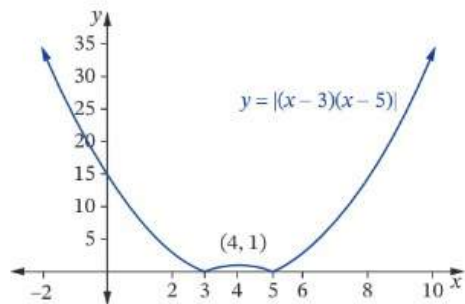
e i



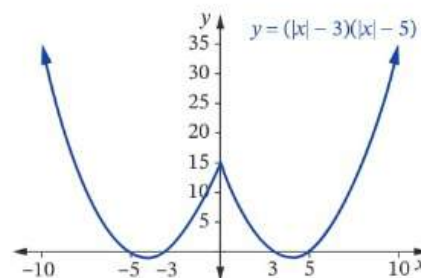
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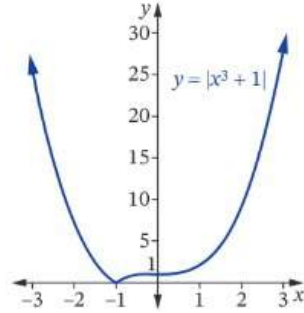
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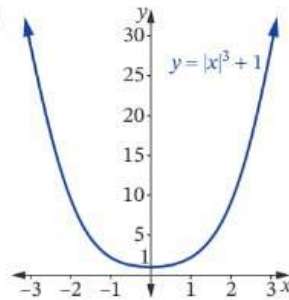
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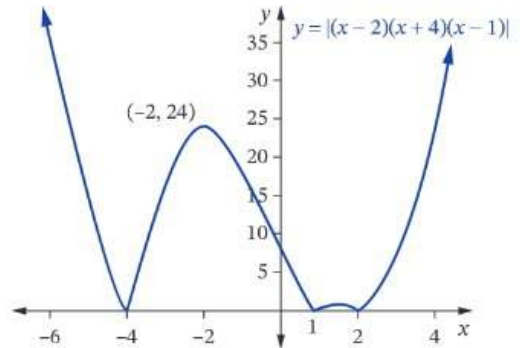
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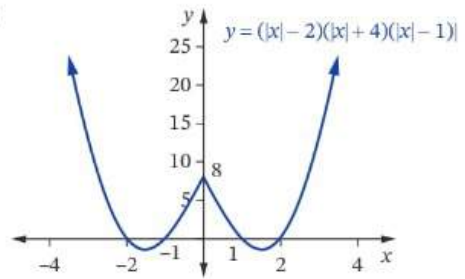
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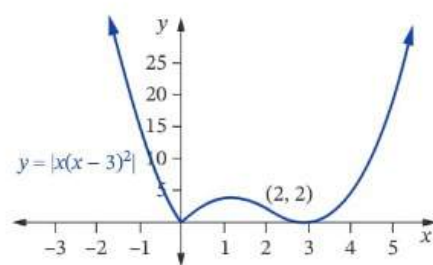
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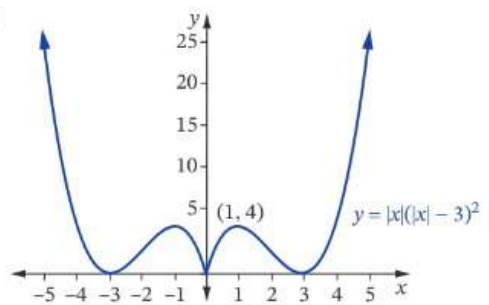
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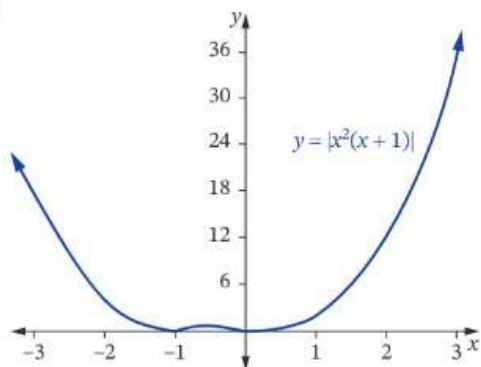
i i



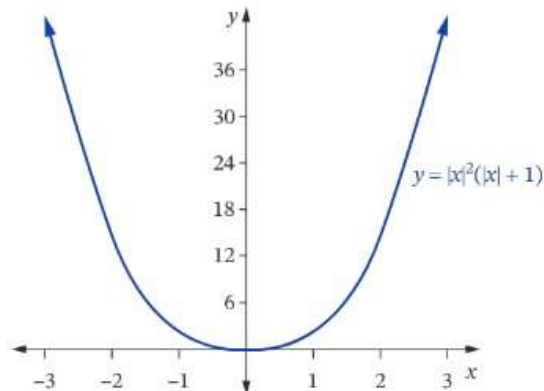
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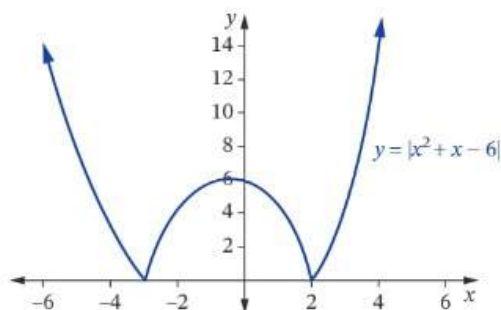
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ii

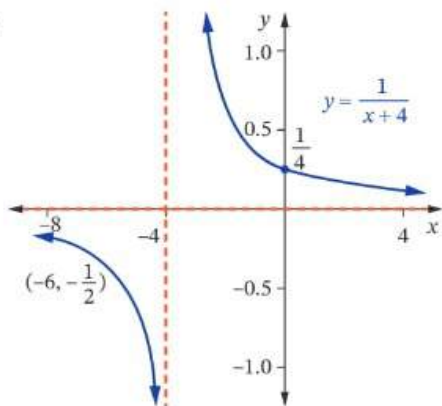


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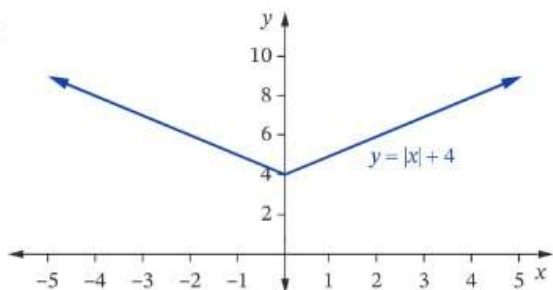


domain: all real x , range: $[0, \infty)$

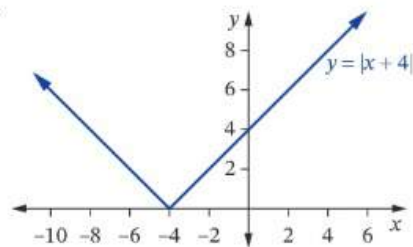
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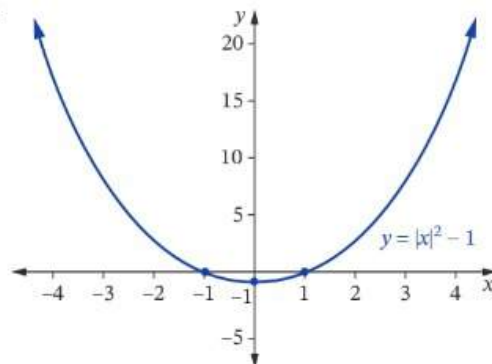
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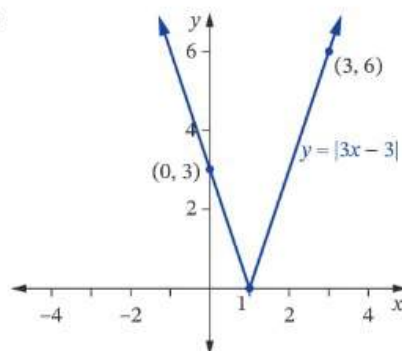
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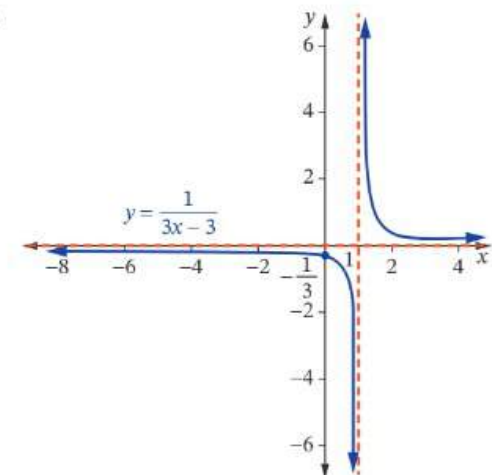
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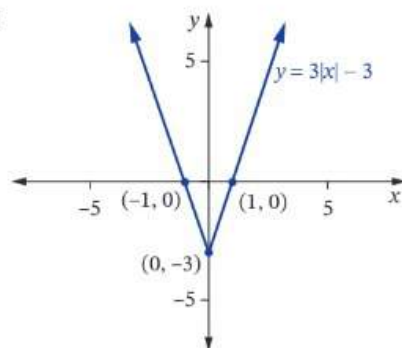
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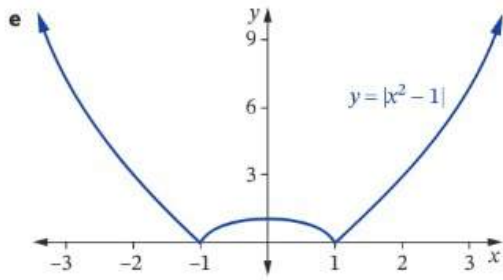


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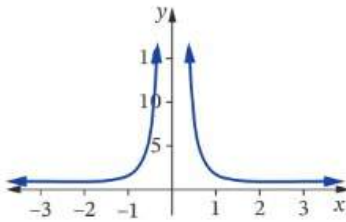


d

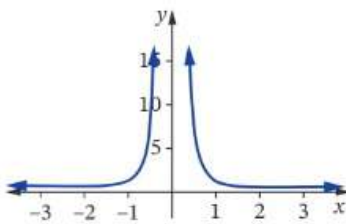




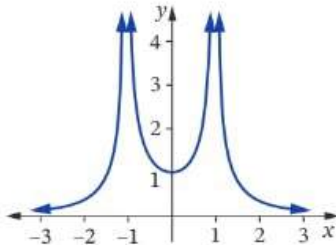
5 a i $y = |f(x)|$



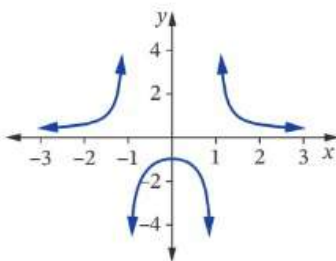
ii $y = f(|x|)$



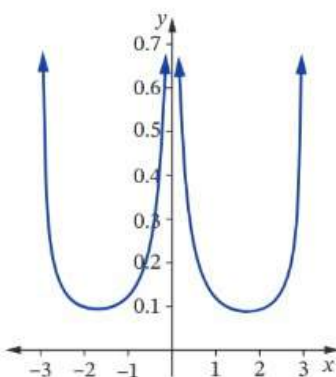
b i $y = |f(x)|$



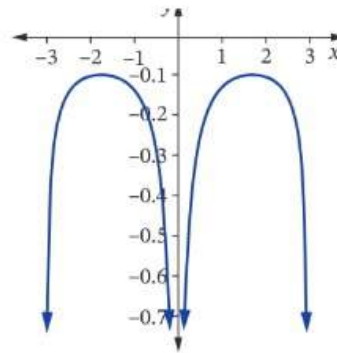
ii $y = f(|x|)$



c i $y = |f(x)|$

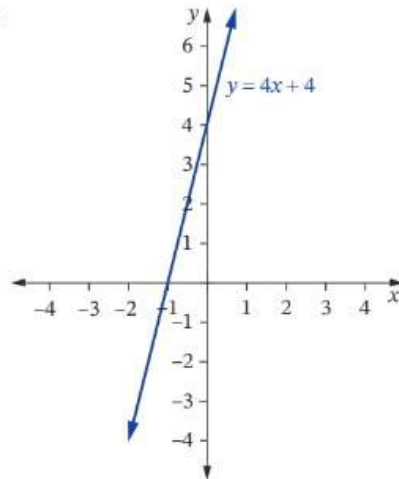


ii $y = f(|x|)$

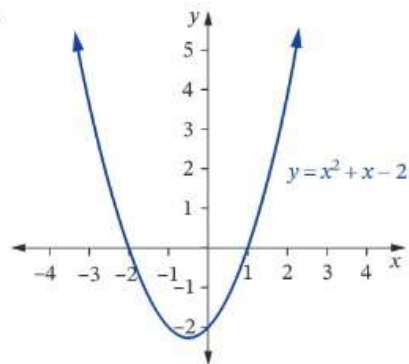


Exercise 6.09

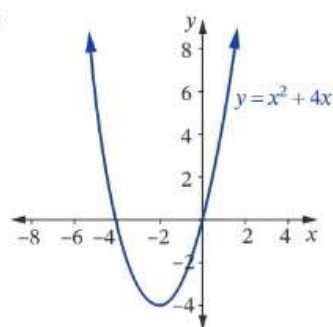
1 a

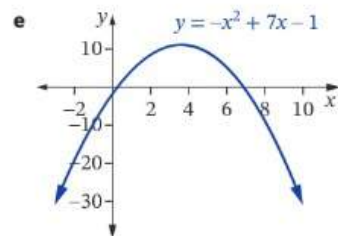
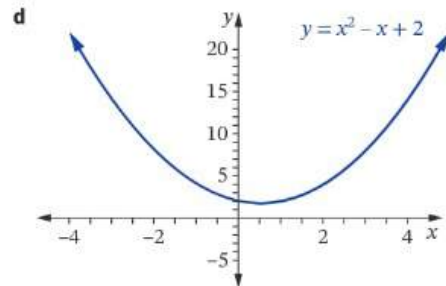
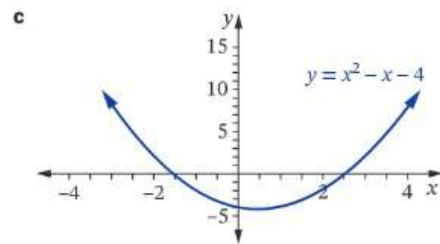
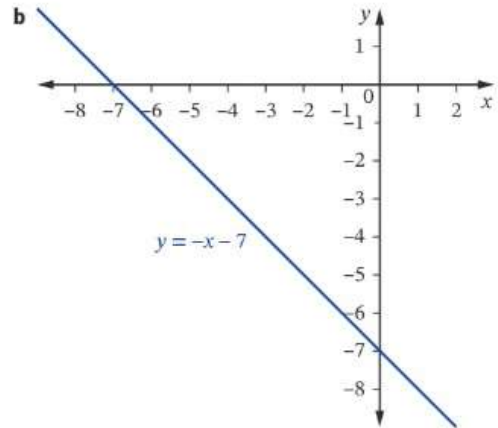
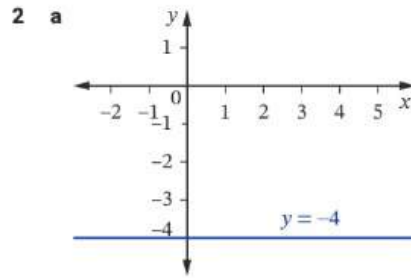
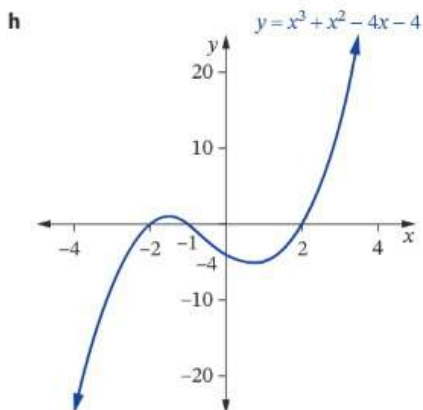
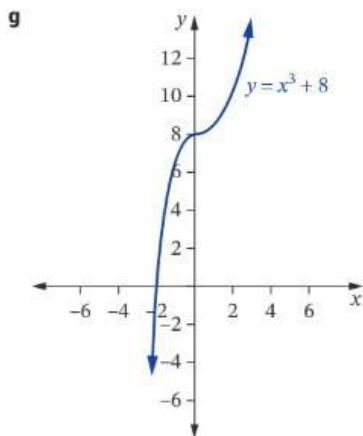
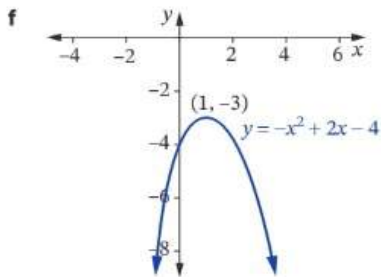
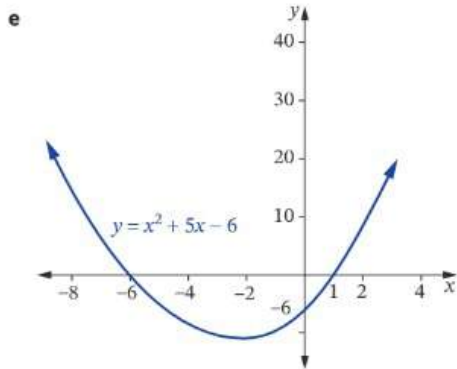
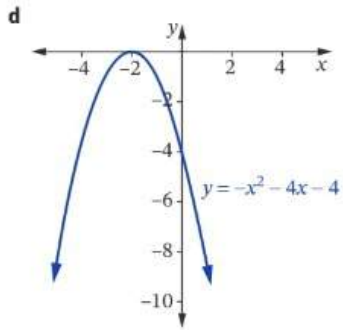


b

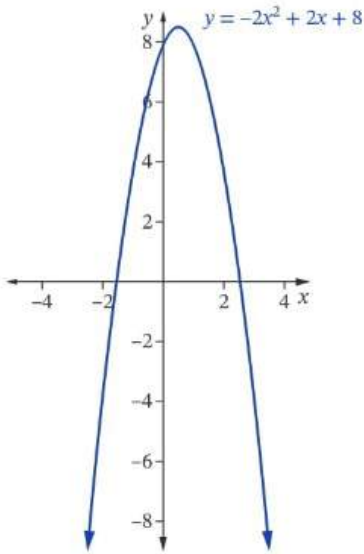


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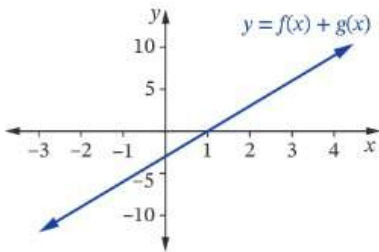




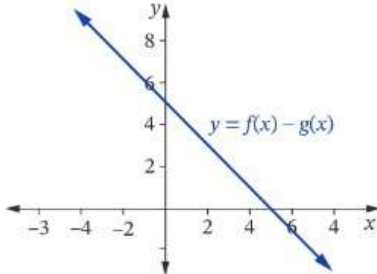
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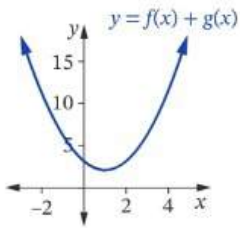
3 a i



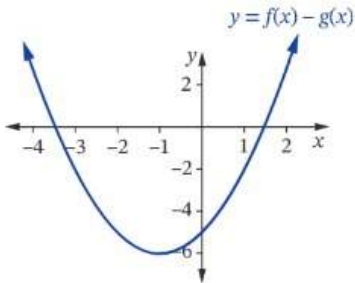
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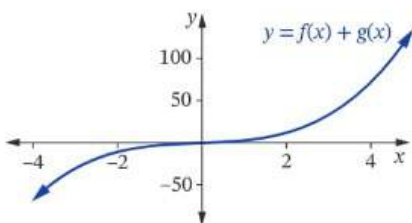
b i



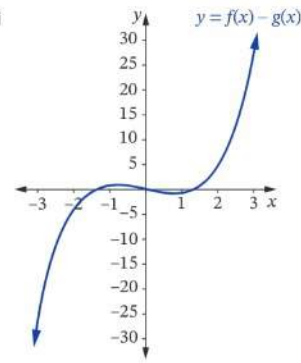
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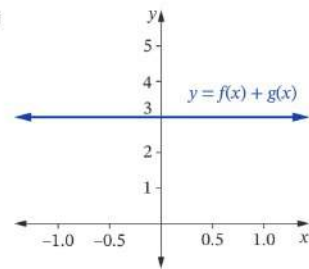
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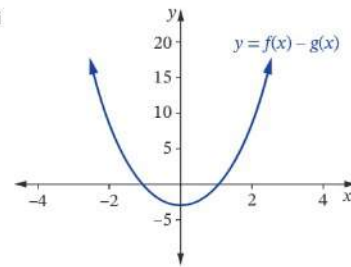
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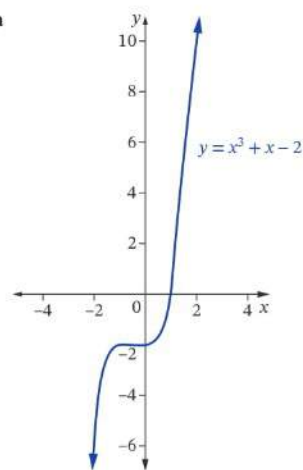
d i



ii

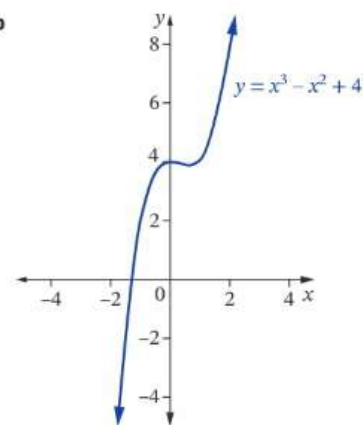


4 a



domain: all real x , range: all real y

b



domain: all real x , range: all real y